Life tables in R using the `tidyverse`

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Life tables are a fundamental tool in demography. A life table describes the mortality experiences for a certain population. Usually a life table is composed of sets of values showing the mortality experience of a hypothetical group of infants born at the same time and subject throughout their lifetime to the specific mortality rates of a given year. Life tables are how we calculate **life expectancy**, probably one of the most common mortality summary measures. They are useful to compare populations and also tells us something about the implied stationary population. Each column refers to a different measure of survivorship. There are different ways of describing survivorship; for example, probability still alive, life expectancy, etc, so a life table has many different columns.

This module explains the main columns of a lifetable and demonstrates how to construct a lifetable in R using the `tidyverse` syntax.

Let's load in the packages we need:

```
library(tidyverse) # data manipulation and ggplot functions
library(kableExtra) # format tables
library(janitor) # to easily clean up column names
```

Columns of the lifetable

Survivorship l_x

Every row of a life table refers to a different age or age group: if the latter, the table is referred to as an **abridged** life table. We will define x to be age and n to be the length of the interval.

A usual place to start is survivorship, l_x, which is defined as the number of people still left alive at age x. The value of l_0 is the starting size of the population, and is called the **radix**. In practice, the radix is usually equal to 1, 100, or 100,000. If $l_0 = 1$ then then l_x is a probability of survival to age x. Note that for now we are implicitly assuming this l_0 relates to a **cohort** of people moving through time, so the life table documents **cohort** mortality. However, later on we will look at period mortality.

Here's the estimated l_x values for females in Ontario in 2015. The data are from the Canadian Human Mortality Database. Here, the radix is 100,000. By age 110, out of the original population of 100,000, it is estimated that 26 will survive.

```
# Download, cache and read the Ontario female survivorship data from period life
tables for females
dir.create("data-cache", showWarnings = FALSE)
fem_lt_url <- "http://www.prdh.umontreal.ca/BDLC/data/ont/fltper_5x5.txt"
fem_lt_file_path <- file.path("data-cache", basename(fem_lt_url))

suppressWarnings(try(
  utils::download.file(fem_lt_url, fem_lt_file_path, mode = "wb", quiet = TRUE),
```

```
  silent = TRUE
))

lt <- tryCatch(
  readr::read_table(fem_lt_url, skip = 2),
  error = function(e) readr::read_table(fem_lt_file_path, skip = 2)
)
```

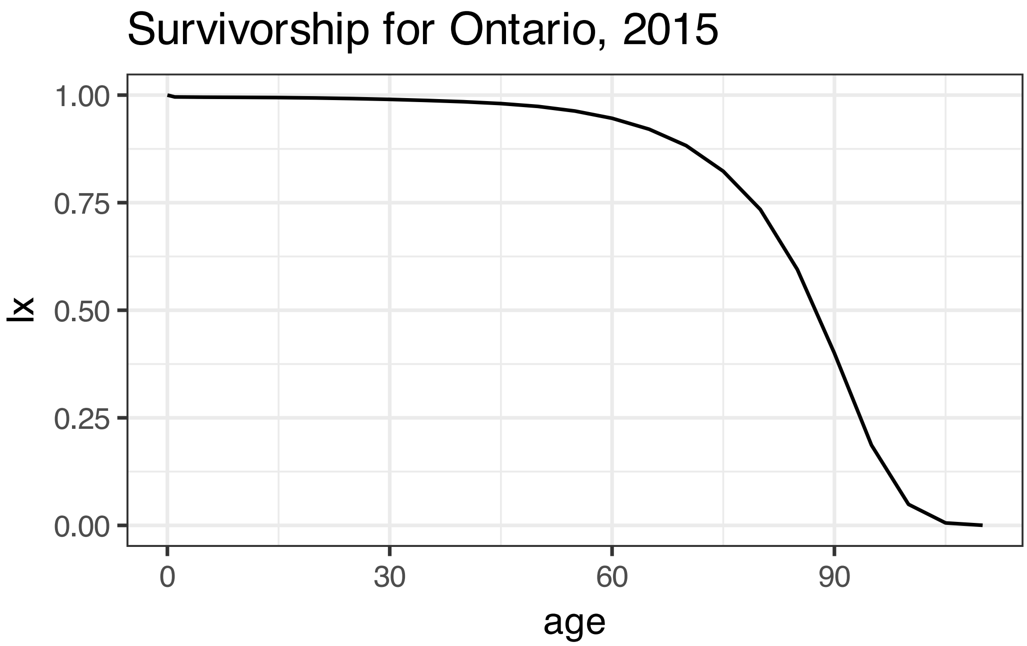
```
# Filter the full table to 2015 female survivorship data, define 5-year age groups
lt <- lt |>
  filter(Year == "2015-2019") |>
  mutate(
    x = c(0, 1, seq(5, 110, by = 5)),
    n = lead(x, default = Inf) - x
  )

# let's look at the data
lt |>
  select(x, n, lx) |>
  kable()
```

x	n	lx
0	1	100000
1	4	99565
5	5	99511
10	5	99474
15	5	99426
20	5	99330
25	5	99187
30	5	99001
35	5	98752
40	5	98453
45	5	98027
50	5	97370
55	5	96290
60	5	94610

x	n	lx
65	5	92074
70	5	88260
75	5	82307
80	5	73420
85	5	59467
90	5	40006
95	5	18659
100	5	4888
105	5	569
110	Inf	26

Let's also plot the l_x and divide through by 100,000 so the l_x can be interpreted as the proportion of the population surviving at age x.

```
lt |>
  mutate(lx = lx / 100000) |>
  ggplot(aes(x, lx)) +
  geom_line() +
  xlab("age") +
  theme_bw(base_size = 14) +
  ggtitle("Survivorship for Ontario, 2015")
```



Deaths ${}_nd_x$

The next column, ${}_nd_x$, is the number of deaths between ages x and $x + n$. Note that we using the 'duration-age' notation, because it refers to deaths over an interval. In contrast, l_x refers to survivors at a certain age x.

By definition, the number of deaths over an interval must be the number of survivors at the start of the interval, minus the number of survivors at the end, i.e.

$$ {}_nd_x = l_x - l_{x+n}. $$

Let's look at the estimated ${}_nd_x$ values for Ontario in 2015:

```
lt |>
  select(x, n, lx, dx) |>
  kable()
```

x	n	lx	dx
0	1	100000	435
1	4	99565	54
5	5	99511	37
10	5	99474	48
15	5	99426	96

x	n	lx	dx
20	5	99330	143
25	5	99187	185
30	5	99001	249
35	5	98752	299
40	5	98453	427
45	5	98027	657
50	5	97370	1079
55	5	96290	1680
60	5	94610	2536
65	5	92074	3814
70	5	88260	5954
75	5	82307	8887
80	5	73420	13953
85	5	59467	19462
90	5	40006	21347
95	5	18659	13771
100	5	4888	4320
105	5	569	543
110	Inf	26	26

Notice the structure of the life table in terms of where the the values of l_x and ${}_nd_x$ line up within the rows. l_x always starts at the radix, so the first row represents the total population before any deaths. The first row of ${}_nd_x$ represents the deaths in the first interval. So the second row of l_x is equal to the previous row of l_x minus the previous row of ${}_nd_x$, and so on. Note also the last interval: everyone who survived to the last age group must die.[1]

Probability of dying, ${}_nq_x$, and of surviving, ${}_np_x$

The next column, ${}_nq_x$ is the probability of dying between ages x and $x + n$. Note that this is a conditional probability, so it's the probability of dying in that interval given you survived to age x. ${}_nq_x$ can be calculated as

$$ {}_nq_x = \frac{{}_nd_x}{l_x} $$

[1] *memento mori.*

The complement of ${}_nq_x$ is the probability of survival, ${}_np_x$

$$ {}_np_x = 1 - {}_nq_x $$

Again, this is a conditional probability, so it's the probability of survival between ages x and $x+n$ given you survived to age x. Given the relationship for ${}_nq_x$ and ${}_nd_x$, we can also calculate ${}_np_x$ as

$$ {}_np_x = 1 - {}_nq_x = 1 - \frac{{}_nd_x}{l_x} = \frac{l_x - l_x + l_{x+n}}{l_x} = \frac{l_{x+n}}{l_x} $$

i.e. the probability of survival is the ratio of the the survivors at the end and start of the interval.

Looking again at the data for Ontario in 2015, notice the probability of death in the last age group is 1, because again, everyone must die eventually.

```
lt |>
  mutate(px = 1 - qx) |>
  select(x, n, lx, dx, qx, px) |>
  kable()
```

x	n	lx	dx	qx	px
0	1	100000	435	0.00435	0.99565
1	4	99565	54	0.00055	0.99945
5	5	99511	37	0.00037	0.99963
10	5	99474	48	0.00049	0.99951
15	5	99426	96	0.00097	0.99903
20	5	99330	143	0.00144	0.99856
25	5	99187	185	0.00187	0.99813
30	5	99001	249	0.00252	0.99748
35	5	98752	299	0.00303	0.99697
40	5	98453	427	0.00433	0.99567
45	5	98027	657	0.00670	0.99330
50	5	97370	1079	0.01109	0.98891
55	5	96290	1680	0.01745	0.98255
60	5	94610	2536	0.02680	0.97320
65	5	92074	3814	0.04142	0.95858
70	5	88260	5954	0.06745	0.93255
75	5	82307	8887	0.10797	0.89203

x	n	lx	dx	qx	px
80	5	73420	13953	0.19004	0.80996
85	5	59467	19462	0.32727	0.67273
90	5	40006	21347	0.53359	0.46641
95	5	18659	13771	0.73802	0.26198
100	5	4888	4320	0.88370	0.11630
105	5	569	543	0.95430	0.04570
110	Inf	26	26	1.00000	0.00000

Average years lived, ${}_na_x$

${}_na_x$ is the number of years lived by those who died between ages x and $x+n$. So for example, if ${}_1a_0 = 0.25$, then for that population, those infants who died in the first year on average died after 0.25 years = 3 months. To calculate the exact value for ${}_na_x$ requires a lot of data: you would need to know the exact lengths of life for each individual in the cohort. Approximations to ${}_na_x$ are discussed below in the period life table section.

Person-years lived, ${}_nL_x$

${}_nL_x$ is the number of person-years lived between ages x and $x+n$. The total number of person-years lived (PYL) in an interval is the sum of

1. the PYL by those who survived and
2. the PYL by those who died in the interval.

The first piece is just the interval length, n multiplied by the number of survivors at the end of the intervals, l_{x+n}. The second piece is the average time spent alive in the interval by those who died, ${}_na_x$ multiplied by the number of people who died in the interval, ${}_nd_x$. So

Note for the last interval there are no survivors, so ${}_\infty L_x = {}_\infty a_x \cdot {}_\infty d_x$.

Adding ${}_na_x$ and ${}_nL_x$ to the Ontario life table:

```
lt |>
  select(x, n, lx, dx, ax, Lx) |>
  kable()
```

x	n	lx	dx	ax	Lx
0	1	100000	435	0.14	99626
1	4	99565	54	1.61	398131
5	5	99511	37	2.34	497456
10	5	99474	48	2.78	497262

x	n	lx	dx	ax	Lx
15	5	99426	96	2.90	496926
20	5	99330	143	2.60	496306
25	5	99187	185	2.53	495475
30	5	99001	249	2.57	494401
35	5	98752	299	2.62	493050
40	5	98453	427	2.68	491278
45	5	98027	657	2.69	488617
50	5	97370	1079	2.72	484383
55	5	96290	1680	2.68	477551
60	5	94610	2536	2.68	467156
65	5	92074	3814	2.67	451498
70	5	88260	5954	2.67	427446
75	5	82307	8887	2.67	390806
80	5	73420	13953	2.68	334729
85	5	59467	19462	2.61	250883
90	5	40006	21347	2.44	145463
95	5	18659	13771	2.19	54585
100	5	4888	4320	1.86	10886
105	5	569	543	1.54	965
110	Inf	26	26	1.44	37

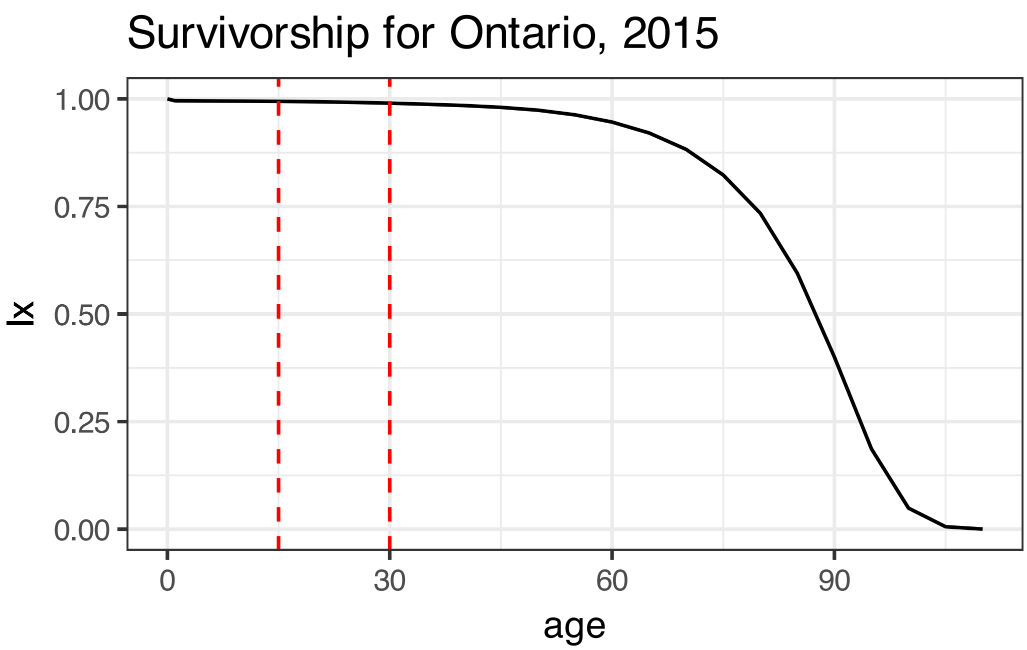
Figure 2 in article: ${}_nL_x$ graphically and the relationship with l_x

${}_nL_x$ is essentially the number of survivors times the average length of time they survived in a particular interval. For a given radix l_0 and interval length n, the maximum ${}_nL_x$ could be is $l_0 \cdot n$, if everyone survived.

How does ${}_nL_x$ relate to l_x? It is the area under the l_x curve for the interval $[x, x+n]$, as illustrated below by the red dashed lines, for ${}_{15}L_{15}$. It may help to think about the units here: ${}_nL_x$ has units person-years. l_x has units of persons. The x-axis on the graph below has units years.

```
lt |>
  mutate(lx = lx / 100000) |>
  ggplot(aes(x, lx)) +
  geom_line() +
```

```
xlab("age") +
theme_bw(base_size = 14) +
geom_vline(xintercept = 15, lty = 2, color = "red") +
geom_vline(xintercept = 30, lty = 2, color = "red") +
ggtitle("Survivorship for Ontario, 2015")
```



With this in mind, we can represent ${}_nL_x$ in continuous form as

$$ {}_nL_x = \frac{1}{l_0} \int_x^{x+n} l_x dx $$

In practice we usually have to calculate ${}_nL_x$ in the discrete form, but it's often useful to think about it in continuous form, i.e. the area under the survivorship curve.

Person-years lived above age x, T_x

Whereas ${}_nL_x$ is the person-years lived in a specific interval, T_x is the person-years lived above a specific age x (so notice it does not have the duration/age notation). It is defined as the sum of the relevant ${}_nL_x$:

In a similar fashion to ${}_nL_x$, T_x can be thought of as the area under the l_x curve above age x. In addition, We can write T_x in continuous form as

$$ T_x = \frac{1}{l_0} \int_x^{\infty} l_x dx $$

Life expectancy at age x, e_x

The final column we will introduce for now is probably the most well-known: e_x, the average number of remaining years of life for those who reach age x, or the **life expectancy** at age x. Note that the 'expectancy' terminology is related to the expected value in the statistical sense. e_x is calculated as

We can do a quick check of the units here to make sure it makes sense: T_x has units person-years, l_x has units persons, so e_x has units of years.

You are probably most familiar with life expectancy at birth, e_0. Note that it is again a conditional measure, that is, it's the average number of remaining years **given** a person has already survived to age x. As such, the value of e_x need not decrease monotonically over age. In practice it usually does, unless infant mortality is relatively high.

The filled-in life table with all columns discussed:

```
lt |>
  select(x, n, lx, dx, ax, Lx, Tx, ex) |>
  kable()
```

x	n	lx	dx	ax	Lx	Tx	ex
0	1	100000	435	0.14	99626	8444919	84.45
1	4	99565	54	1.61	398131	8345293	83.82
5	5	99511	37	2.34	497456	7947162	79.86
10	5	99474	48	2.78	497262	7449706	74.89
15	5	99426	96	2.90	496926	6952444	69.93
20	5	99330	143	2.60	496306	6455517	64.99
25	5	99187	185	2.53	495475	5959212	60.08
30	5	99001	249	2.57	494401	5463736	55.19
35	5	98752	299	2.62	493050	4969335	50.32
40	5	98453	427	2.68	491278	4476285	45.47
45	5	98027	657	2.69	488617	3985007	40.65
50	5	97370	1079	2.72	484383	3496389	35.91
55	5	96290	1680	2.68	477551	3012006	31.28
60	5	94610	2536	2.68	467156	2534455	26.79
65	5	92074	3814	2.67	451498	2067299	22.45
70	5	88260	5954	2.67	427446	1615801	18.31
75	5	82307	8887	2.67	390806	1188355	14.44

x	n	lx	dx	ax	Lx	Tx	ex
80	5	73420	13953	2.68	334729	797548	10.86
85	5	59467	19462	2.61	250883	462819	7.78
90	5	40006	21347	2.44	145463	211936	5.30
95	5	18659	13771	2.19	54585	66473	3.56
100	5	4888	4320	1.86	10886	11888	2.43
105	5	569	543	1.54	965	1002	1.76
110	Inf	26	26	1.44	37	37	1.44

Period life tables

A life table as defined above refers to tracking the mortality of a **cohort** of people as they age. However, it is often not practical or useful just to consider the mortality of a cohort, because in order to to build a complete table, we have to wait to observe everyone in the cohort die. So for the 1990 birth cohort, for example, we would probably have to wait around until at least 2090 before a reasonable cohort life table could be built. This is not very useful to study current mortality conditions.

In addition to constructing cohort life tables, we can construct **period** life tables. They refer to the period in the sense that they are constructed using mortality conditions in a particular period. This means the life table refers to a **synthetic cohort**, a hypothetical group of people that experience the mortality conditions of the period of interest throughout their entire life. Why is this hypothetical and potentially unrealistic? Because mortality conditions change over time (and in general, are getting better). So for example, if I live until I'm 70, the mortality conditions I am subject to in the future are likely to be different to the mortality conditions that a current 70-year-old is being subjected to. However, period life tables are still useful to compare mortality outcomes for different populations in a more up-to-date way.

Construction from period mortality rates

The key to constructing period life tables is converting the observed period mortality rates ${}_nM_x$ to probabilities of death, ${}_nq_x$. The mortality rate is the number deaths divided by person years lived, so in life table notation this is:

$${}_nM_x = \frac{{}_nd_x}{{}_nL_x}$$

We can use the following **${}_nM_x$ to ${}_nq_x$ conversion formula** to get the ${}_nq_x$ column, after which all other columns can be derived based on the relationships discussed above (and choosing a radix). The formula is:

$${}_nq_x = \frac{n \cdot {}_nM_x}{1 + (n - {}_na_x) \cdot {}_nM_x}$$

How did this come about? By rewriting

$$ {}_nL_x = n \cdot l_{x+n} + {}_na_x \cdot {}_nd_x = n(l_n - {}_nd_x) + {}_na_x \cdot {}_nd_x $$

rearranging to get l_x and rewriting ${}_nq_x$ with this denominator.

Getting values for ${}_na_x$

The conversion formula requires information only on period mortality rates and values of the average number of years lived for those who died, ${}_na_x$. How do we get these values? As mentioned above, the data required to calculate ${}_na_x$ exactly is usually not available, so we need to approximate it somehow. Preston, Heuveline and Guilot (2000) has a good overview of the options here (section 3.2, page 44), but the most common and easiest approach is, for **most** age groups, assume

$$ {}_na_x = n/2 $$

that is, on average those who die, die half-way through the interval. So for an abridged life table with five-year intervals, ${}_na_x = 2.5$. This assumption is fine for most age groups, except for the very young and very old ages.

At younger ages, typically in abridged life tables the first five years is split out into the first year, and years 1-5, so we need values for ${}_1a_0$ and ${}_4a_1$. For mortality at younger ages, we expected comparatively more deaths to occur at the start of the interval, so ${}_na_x < n/2$. The following two approximations are common, and used e.g. by Wachter in Essential Demographic Methods (2014):

$$ {}_1a_0 = 0.07 + 1.7{}_1M_0 $$

and

$$ {}_4a_1 = 1.5 $$

The equation for ${}_1a_0$ says that infants who die on average die at about 1 month plus a bit, where the bit depends on the over level of infant mortality. This is based on the observation that as infant mortality declines, infants that are dying are more likely to die from pre-existing conditions rather than exogenous factors, so deaths occur relatively early.

For the last age group, we can assume ${}_na_x$ is the inverse of the mortality rate:

$$ {}_\infty a_\omega = 1/{}_\infty M_\omega $$

where ${}_\infty M_\omega$ is the age-specific mortality rate for the last age interval; ω refers to the last age. The smaller the interval of the open-ended age group, the better approximation.

Interpretation of period life table measures

As mentioned at the start of this section, period life tables are constructed from a synthetic cohort of people that hypothetically would go through life experiencing the age-specific mortality conditions of the current period. Period life table measures, and in particular, period life expectancy at birth, are the most commonly published and discussed. However, period measures of life expectancy are often misinterpreted. The technical definition is the expected number of years of live for a newborn who would be subject to the current mortality conditions for their entire life. But life expectancy is usually just talked about as 'how long you're expected to live for'. Part of the confusion comes from the name, but the 'expected' part refers to the fact that it is an expected value in the statistical sense; in particular, $E[T_x] = e_x$.

R: Make your own life table

Let's make a period life table for females in Quebec in 2015 using data from the Canadian Human Mortality Database. Read in the data directly from the website, and filter out what we need:

```
# Download, cache, and read the deaths data
deaths_url <- "https://www.prdh.umontreal.ca/BDLC/data/ont/Deaths_5x1.txt"
deaths_file_path <- file.path("data-cache", basename(deaths_url))
suppressWarnings(try(
  utils::download.file(deaths_url, deaths_file_path, mode = "wb", quiet = TRUE),
  silent = TRUE
))
deaths <- tryCatch(
  readr::read_table(deaths_url, skip = 2, col_types = "ccddd"),
  error = function(e) readr::read_table(deaths_file_path, skip = 2, col_types
= "ccddd")
)
```

```
# Download, cache, and read the population data
pop_url <- "https://www.prdh.umontreal.ca/BDLC/data/ont/Population5.txt"
pop_file_path <- file.path("data-cache", basename(pop_url))
suppressWarnings(try(
  utils::download.file(pop_url, pop_file_path, mode = "wb", quiet = TRUE),
  silent = TRUE
))
pop <- tryCatch(
  readr::read_table(pop_url, skip = 2, col_types = "ccddd"),
  error = function(e) readr::read_table(pop_file_path, skip = 2, col_types
= "ccddd")
)
```

```
# Clean names and select columns of interest in each dataframe
deaths <- deaths |>
  mutate(year = as.numeric(substr(Year, 1, 4))) |>
  select(year, Age, Total) |>
  clean_names() |>
  rename(D = total)
pop <- pop |>
```

```
  mutate(year = as.numeric(substr(Year, 1, 4))) |>
  select(year, Age, Total) |>
  clean_names() |>
  rename(N = total)
```

```
# join population and deaths
d <- deaths |>
  left_join(pop, by = c("year", "age"))
```

The `age` column is a character, so let's make an age `x` and interval length `n` column. We also aggregate ages 95+ because older ages have varying data quality

```
d <- d |>
  mutate(x = as.numeric(str_remove(age, "-.*|\\+"))) |>
  mutate(x = ifelse(x >= 95, 95, x)) |>
  group_by(year, x) |>
  summarise(
    D = sum(D),
    N = sum(N),
    .groups = "drop"
  ) |>
  mutate(
    age = case_when(
      x == 0 ~ "0",
      x == 1 ~ "1-4",
      x == 95 ~ "95+",
      TRUE ~ paste0(x, "-", x + 4)
    )
  ) |>
  select(year, age, x, N, D)
```

Now we can use `tidyverse` to calculate the columns in the life table, based on the equations presented in previous sections. I set the radix l_0 to be one and filter to just include the year 2015. This code makes use of the `case_when` function, which allows to define different values of ${}_na_x$ based on age group. Formulas for other columns are based on equations stated above. The formula for T_x is implemented by first reversing the ${}_nL_x$ column, taking the cumulative sum, and then reversing the result.

```
lt_2015 <- d |>
  filter(year == 2015) |>
  mutate(
    n = lead(x, default = Inf) - x,
```

```
    Mx = D / N,
    ax = case_when(
      x == 0 ~ 0.07 + 1.7 * Mx,
      x == 1 ~ 1.5,
      x == 95 ~ 1 / Mx,
      TRUE ~ 2.5
    ),
    qx = case_when(
      x == 95 ~ 1, # set to 1 for open interval
      TRUE ~ n * Mx / (1 + (n - ax) * Mx)
    ),
    px = 1 - qx,
    lx = lag(cumprod(px), default = 1),
    dx = lx - lead(lx, default = 0),
    Lx = case_when(
      x == 95 ~ lx / Mx,
      TRUE ~ n * lead(lx, default = 0) + (ax * dx)
    ),
    Tx = rev(cumsum(rev(Lx))),
    ex = Tx / lx
  )
```

Table 2: Period life table for females in Quebec in 2015

```
lt_2015 |>
  select(Age = age, x, n, Mx, ax, qx, px, lx, dx, Lx, Tx, ex) |>
  # Padding Age with non-breaking spaces so it doesn't overflow into x
  mutate(Age = paste0(Age, strrep("\u00a0", 2))) |>
  kable(
    booktabs = TRUE,
    digits = c(NA, 0, 0, 6, 3, 6, 6, 6, 6, 3, 3, 3)
  )
```

Age	x	n	Mx	ax	qx	px	lx	dx	Lx	Tx	ex
0	0	1	0.004424	0.078	0.004406	0.995594	1.000000	0.004406	0.996	82.314	82.314
1-4	1	4	0.000160	1.500	0.000640	0.999360	0.995594	0.000637	3.981	81.318	81.677
5-9	5	5	0.000072	2.500	0.000360	0.999640	0.994957	0.000358	4.974	77.337	77.729
10-14	10	5	0.000113	2.500	0.000563	0.999437	0.994599	0.000560	4.972	72.363	72.756
15-19	15	5	0.000247	2.500	0.001232	0.998768	0.994039	0.001225	4.967	67.391	67.795
20-24	20	5	0.000461	2.500	0.002302	0.997698	0.992814	0.002285	4.958	62.424	62.876

Age	x	n	Mx	ax	qx	px	lx	dx	Lx	Tx	ex
25-29	25	5	0.000523	2.500	0.002609	0.997391	0.990529	0.002585	4.946	57.466	58.015
30-34	30	5	0.000621	2.500	0.003100	0.996900	0.987944	0.003063	4.932	52.520	53.161
35-39	35	5	0.000728	2.500	0.003636	0.996364	0.984881	0.003581	4.915	47.588	48.318
40-44	40	5	0.001081	2.500	0.005391	0.994609	0.981301	0.005290	4.893	42.672	43.485
45-49	45	5	0.001646	2.500	0.008198	0.991802	0.976010	0.008001	4.860	37.779	38.707
50-54	50	5	0.002846	2.500	0.014129	0.985871	0.968009	0.013677	4.806	32.919	34.007
55-59	55	5	0.004644	2.500	0.022956	0.977044	0.954332	0.021908	4.717	28.113	29.458
60-64	60	5	0.007311	2.500	0.035896	0.964104	0.932425	0.033471	4.578	23.396	25.092
65-69	65	5	0.010918	2.500	0.053141	0.946859	0.898954	0.047772	4.375	18.818	20.933
70-74	70	5	0.017563	2.500	0.084121	0.915879	0.851182	0.071602	4.077	14.442	16.967
75-79	75	5	0.028239	2.500	0.131884	0.868116	0.779580	0.102814	3.641	10.365	13.296
80-84	80	5	0.051243	2.500	0.227120	0.772880	0.676766	0.153707	3.000	6.725	9.936
85-89	85	5	0.092581	2.500	0.375902	0.624098	0.523058	0.196619	2.124	3.725	7.122
90-94	90	5	0.166120	2.500	0.586873	0.413127	0.326440	0.191579	1.153	1.601	4.905
95+	95	Inf	0.301053	3.322	1.000000	0.000000	0.134861	0.134861	0.448	0.448	3.322